



Carnegie Mellon University
Master of
Software Engineering

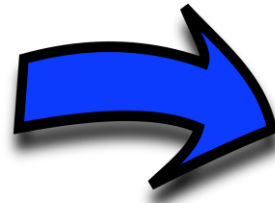
Architecture – motivation and definition

LG Architecture Training Program

Paulo Merson

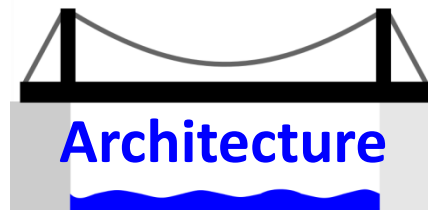
Motivation for Software Architecture ₍₁₎

The challenge of creating software



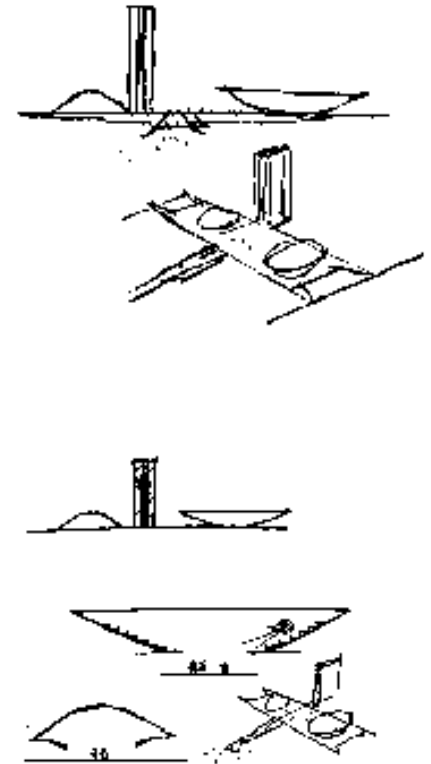
Motivation for Software Architecture ₍₂₎

The role of software architecture



What is software architecture?

- What is *your* definition?
- Concretely, what is the deliverable artifact for a software architecture?



*As the size and complexity of software systems increases, **the design problem goes beyond the algorithms and data structures** of the computation: designing and specifying the **overall system structure** emerges as a new kind of problem. Structural issues include gross organization and global control structure; protocols for communication, synchronization, and data access; assignment of functionality to design elements; composition of design elements; scaling and performance; and **selection among design alternatives**. This is the software architecture level of design.*

[Garlan94]

Mary Shaw



David Garlan



*An architecture is the **set of significant decisions about the organization of a software system**, the selection of the **structural elements and their interfaces** by which the system is composed, together **with their behavior** as specified in the collaborations among those elements, the composition of these structural and behavioral elements into progressively larger subsystems, and the architectural style that guides this organization—these elements and their interfaces, their collaborations, and their composition.*
[Kruchten03]

Philippe Kruchten



*Fundamental concepts or properties of a system in its environment embodied in its **elements, relationships, and in the principles of its design** and evolution.*
[ISO 42010]



Grady Booch ✓
@Grady_Booch



Replying to @notessensei

Software architecture represents the significant design decisions that shape a system, where significant is measured by cost of change.

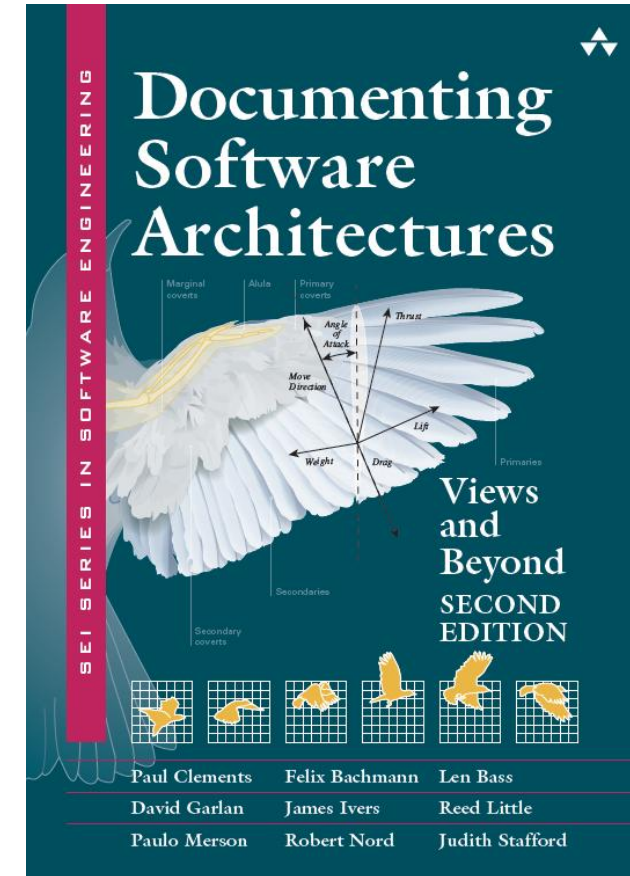
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Software architecture definition

- This is the definition we'll use in this course

*The software architecture of a system is the **set of structures** needed to reason about the system, which comprise **software elements**, **relations among them**, and **properties of both**.*
[Clements10]

- Not only “what”, but also “why”



Which structures?

- A software architecture is composed of many structures, such as:
 - Code units and their decomposition in submodules
 - Code units and their inter-dependencies
 - Services and how they interact at runtime
 - How components are packaged and deployed on the runtime infrastructure
 - Domain model or data model



We'll get
back to this
point in
week 2

Architecture vs design

- *Architecture is design, but not all design is architectural*

- A better separation is:
 - architectural design
 - non-architectural design

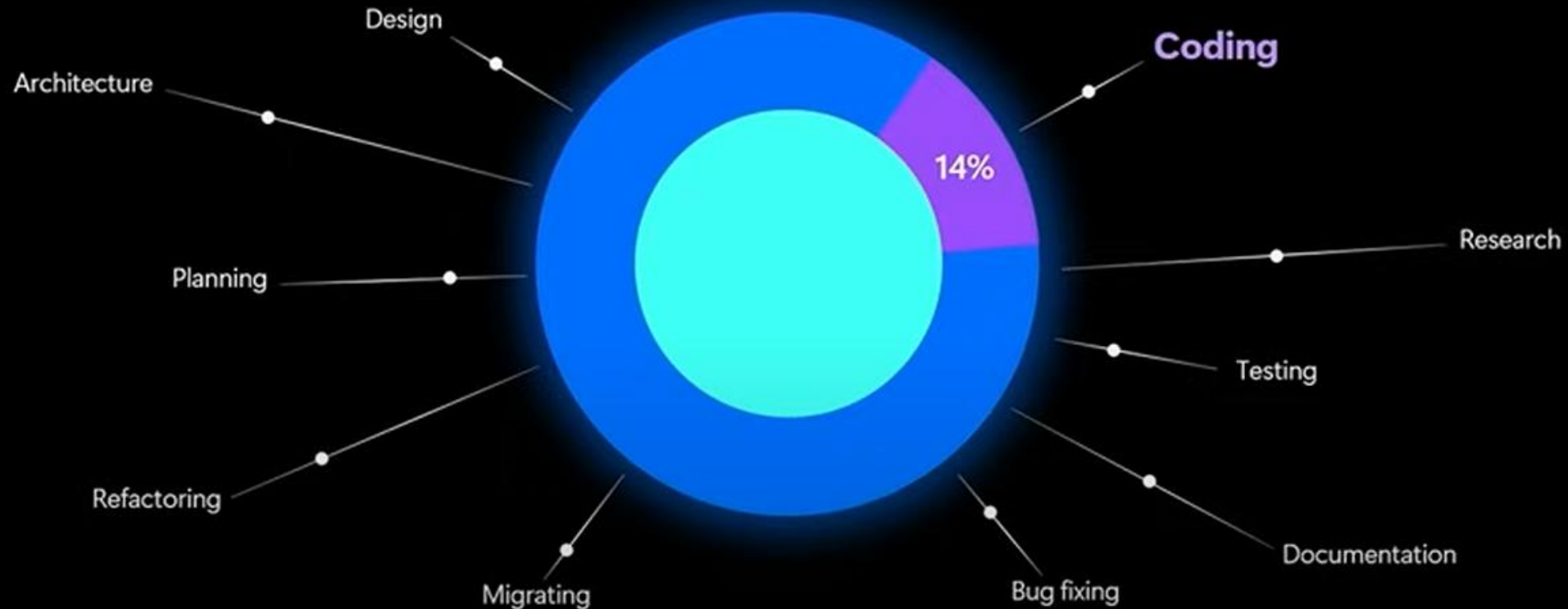
Decisions that affect the ability of the system to address the quality requirements and exhibit the expected behavior

What's the difference between architecture and design?

- The architect draws the line between the two



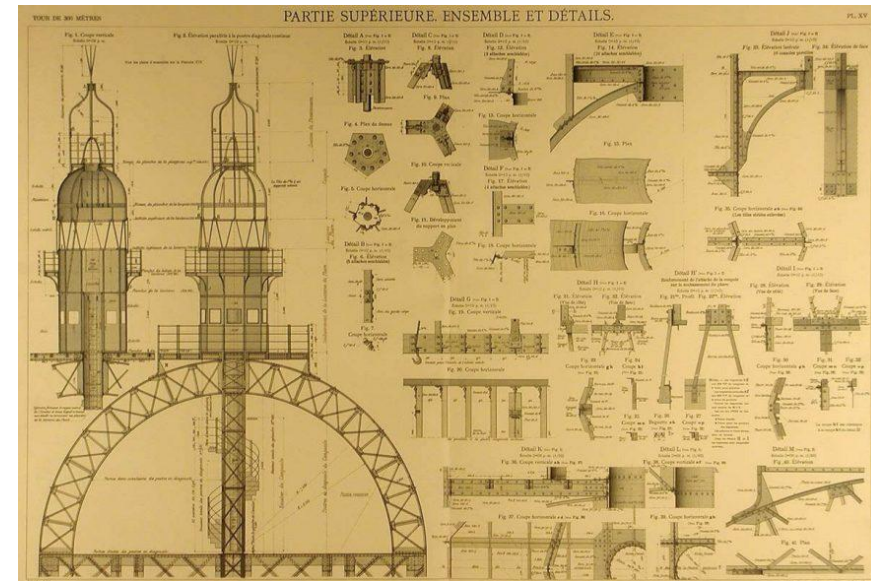
Developers spend only a fraction of their time writing new code



Source: [Reimagining Software Development and DevOps with Agentic AI](#) (Microsoft)

Creativity and knowledge

- A good architecture is
 - the result of consistently and continuously applying principles, techniques and patterns
 - resilient in the face of (inevitable) changes
 - a guide in each phase of software development
 - the result of established and mature architecture knowledge



Uses for software architecture

- It is the blueprint to construct the system
- It helps in project management
 - Work assignment and prioritization
 - Tracking progress
- It is the primary source of quality attributes
 - Enables early assessment of a system's qualities
- It helps in the onboarding of new developers
- It is input to the work of several stakeholders





Questions?

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